INCH-POUND

MIL-T-87984
31 January 1989

MILITARY SPECIFICATION

TRAILER, TANK, FUEL SERVICING, TYPE A-1B

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a trailer with mounted tank for storing, transporting and servicing 600 gallons of all types of aircraft, ground and ground support equipment fuels.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks for a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

VV-F-800	Fuel Oil, Diesel

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMVRS, ROBINS AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2330

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MILITARY

MIL-P-514	Plates, Identification, Instructions and Marking, Blank
MIL-G-3056	Gasoline, Automotive, Combat
MIL-G-5572	Gasoline, Aviation
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-H-6615	Hose Assemblies, Rubber, Fuel and Water, with Reattachable Couplings, Low Temperature
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-P-23377	Primer Coating, Epoxy-Polyamide, Chemical and Solvent Resistant
MIL-T-38219	Turbine Fuel, Low Volatility, JP-7
MIL-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-C-53039	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8
MIL-R-83232	Reel, Static Discharge, Grounding, RMU-2251E.50 Foot with Connector
MIL-C-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydro-Carbon Base, Aircraft
MIL-C-83286	Coating, Urethane Aliphatic Isocyanate for Aerospace Applications

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-12	Abbreviations for Use on drawings and on Specifications Standards and Technical Documents
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications. Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Emission and Susceptability Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics Measurements of Acceptance Tests: Expotential Distribution
MIL-STD-808	Finishes Materials and Processes for Corrosion Prevention and Control in Support Equipment

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MIL-STD-810	Environment Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1179	Lamps, Reflectors and Associated Signaling Equipment for Military Vehicles
MIL-STD-1223	Nontactical Wheeled Vehicle Treatment Painting Identification Marking and Data Plate Standards
MIL-STD-1472	Human Engineering Design Criteria for Military Systems Equipment and Facilities
MIL-STD-1595	Qualification of Aircraft, Missile and Aerospace Fusion Welders
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft

DESIGN HANDBOOK

AIR FORCE SYSTEMS COMMAND (AFSC)

AFSC DH 1-11	Air Transportability

DRAWINGS

MS 27020
MS 27022
MS 27024
MS 27026
MS 27028
MS 27029
MS 51336

(Copies of specifications, standards, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Carrier Safety Regulations (CFR)
Federal Motor Carrier Safety Standards (CFR)

(Application for copies should be addressed to the Department of Transportation. Federal Highway Administration, Washington DC 20591.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 385	Tank Vehicle Design
NFPA 407	Aircraft Fuel Servicing

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J318	Air Brake Gladhand Service and Emergency Line Couplers-Trucks, Truck-Tractors and Trailers
J534	Lubrication Fittings
J536	Hose Clamps
J682	Rear Wheel Splash and Stone Throw Protection
J683	Tire Chain Clearance
J695	Turning Ability and Off Tracking-Motor Vehicles
J1349	Engine Power Test Code-Spark Ignition and Diesel

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale PA 15096.)

THE TIRE AND RIM ASSOCIATION

Yearbook

(Application for copies should be addressed to The Tire and Rim Association, Incorporated, 3200 West Market Street, Akron, OH 44313.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards). The text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design

3.1.1 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

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3.1.2 Materials. Materials shall be as specified herein. Materials not specifically covered by this specification or referenced document shall be suitable in every respect for this type equipment. Except as otherwise specified wood shall not be used.

3.1.2.1 Protective treatment. Materials that are subject to deterioration shall be protected in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack or scale with age or extremes of climatic conditions shall not be used.

3.1.2.2 Dissimilar metals. Unless protected against electrolytic corrosion dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.1.3 Design and construction. The vehicle shall be constructed so that parts do not work loose in service and shall withstand the loads, shocks, vibrations and other conditions incident to operation, shipping and storage with minimum loss of time for maintenance, repair and servicing. The design of the vehicle shall ensure safe, efficient and economical operation. All assemblies, controls and installed equipment shall be located so there is no adverse interference with each other or with the operation and shall be readily accessible for maintenance operation removal and replacement.

3.1.3.1 Reliability. The vehicle including all components shall have a Mean-Time-Between-Failures (MTBF) of not less than 100 hours demonstrated at an 0.9 confidence level. The vehicle shall be designed for a life expectancy of at least 15 years.

3.1.3.2 Maintainability. The vehicle shall be designed and constructed to provide the following:

a. A minimum number of parts consistent with the performance required herein.

b. A minimum amount of time for assembling, disassembling, locating trouble sources and maintaining the vehicle. Where practical components shall be located for rapid inspection and recognition of potential failure.

c. Ease of adjusting, servicing and replacing parts and components.

d. Use of readily available standard tools and equipment for maintenance.

e. Maintenance with a minimum number of tools.

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f. Mean corrective maintenance downtime of no greater than one hour.

g. Maximum corrective maintenance downtime of no greater than four hours.

h. Maximum total downtime to perform all daily scheduled maintenance and servicing tasks of no greater than 35 minutes.

i. The vehicle shall be designed so that one mechanic using common hand tools can drain and refill engine lubricating oil, remove and replace all screens, strainers, hoses and filter elements within one hour.

3.1.3.3 Foolproofness. Where improper installation of an item could cause a malfunction, an asymmetric mounting system shall be provided where practical to ensure proper mounting of the item.

3.1.3.4 Fastening devices. All screws, bolts, nut, pins and other fastening devices shall be properly designed, manufactured and installed with adequate means of preventing loss of torque or adjustment. Cotter, pins, lock washers or nylon patches shall not be used for this purpose. Tapped aluminum threads shall have a minimum thread engagement of at least two times the nominal fastener diameter.

3.1.3.5 Servicing provisions. Drains grease fittings, lines and check points for all components shall be so located that they are readily accessible and do not require special tools for proper servicing.

3.1.3.7 Component protection. All space in which work is performed during operation, service and maintenance of the vehicle shall be free of hazardous protrusions, sharp edges or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized that are of such nature or so located as to be hazardous to personnel shall be guarded or insulated to the extent necessary to eliminate the hazard.

3.1.3.8 Foreign object damage. All loose metal parts such as pins and valve caps shall be securely attached so they cannot become separated from the vehicle.

3.1.3.9 Bonding. Metal parts shall be specifically bonded together to minimize the possibility of sparks.

3.1.4 Environmental conditions. The vehicle shall be capable of satisfactory storage and operation under the following environmental conditions:

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a. Full-time exposure to temperatures ranging from -40°F to +125°F for operation: -60°F to +160°F for storage with proper lubrication.

b. Full-time exposure to relative humidities up to 100 percent.

c. Part-time (five percent of the time) exposure to rain; droplets of size 2.0 to 4.5 mm falling at a rate of 5.2 inches per hour.

d. Part-time (five percent of the time) exposure to salt fog; five percent solution (five parts by weight of salt in 95 parts by weight of water) with fallout of 3.0 ml/80cm2/ hour.

e. Part-time (five percent of the time) exposure to blowing dust; velocity 1750 ft/min. concentration 0.5 g/ft^3.

f. Part-time (five percent of the time) exposure to blowing sand; velocity 5700 ft/min. concentration 0.0773 g/ft^3.

3.1.5 Weight distribution. The weight distribution of the vehicle in any loading condition shall not be unbalanced or improperly distributed. The difference in load between the left and right wheels of any axle shall not exceed 2.5 percent of the total load on the axle.

3.1.6 Body lean. The vehicle shall not lean to either side more than one percent of the body width in either the full loaded or the unloaded conditions. Body lean measurements shall be made at the outermost portion of the vehicle (excluding lights or asymmetrical shapes) directly above the axle: the difference in body height at the axle shall not exceed one percent of the body width at that location.

3.1.7 Electromagnetic interference. The vehicle shall comply with the requirements of MIL-STD-461. Class C1, Group II for electromagnetic interference except that the limits for UM03 (Figure 8-1) shall be relaxed by 20dB.

3.1.8 Equipment operation sound level. The maximum A-weighted sound level produced by the trailer's engine pump assembly shall not exceed 84dBA. The noise level shall be measured at the operator's station adjacent to the controls with the operator standing on the ground. The sound level shall be measured in accordance with 29CFR 1910.95.

3.1.9 Tiedowns. The vehicle shall be equipped with sufficient tiedown points so located that it can be restrained in all directions during air transport. The attachment points shall be designed for use with attachment in the aircraft as described in AFSC DH-1-11. Sufficient clearance shall be maintained around

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all attachment points to allow for restraining chains without interference from any other structure. The tiedown points shall have a minimum clear opening of three inches to allow attachment of a chain with a grabhook.

3.2 Performance

3.2.1 Mobility. The fully loaded trailer shall be capable of being towed by a commercial type truck or military design M35. When in connection with the prime mover the fully loaded trailer shall be capable of performing the following:

a. Operating as required without adversely affecting vehicle operation or spillage of fuels or lubricants while ascending or descending 30 percent grades and/or 30 percent side slopes in both directions.

b. Withstanding the shocks, vibrations and other treatments the trailer may experience while being towed at rated speeds and normal operating conditions.

c. Comply with DOT brake performance requirements for service brakes, emergency brakes and parking brakes.

d. Being towed at a minimum speed of 60 MPH over a dry smooth prepared hard surface. When towed at curb and gross weights in a straight path for a distance of 150 feet the path of the towed chassis shall not deviate more than three inches to either side of the path of the towing vehicle.

e. Being towed over unimproved roads, trails, open fields, rolling hills and cross country terrain at speeds up to 35 MPH.

f. Traverse Belgian Block at a speed of 20 MPH.

h. Negotiate short radius turns under any load condition. When coupled to the towing vehicle and operating in it minimum turning circle the trailer shall follow without cramping or interference.

i. Assuming a 70° turn angle when coupled to the prime mover without interference between vehicles. The angle shall be measured between the longitudinal axis of the trailer and longitudinal axis of the towing vehicle at the lunette.

j. Ford salt or fresh water to not less than a depth of 24 inches without any special servicing or protection before or after fording. No evidence of water leakage or moisture penetration into the sealed wheel bearings shall be permitted.

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3.3 Chassis. The trailer components shall be mounted on a single axle trailer chassis with an adjustable retractable landing gear. All trailer chassis components including frame, axle, springs and bearings shall be designed and constructed to withstand the forces imposed on the fully loaded vehicle.

3.3.1 Towbar. The trailer chassis shall be equipped with a towbar designed for towing by vehicles having a pintle height of 30 to 35 inches and a lunette eye conforming to MS 51336.

3.3.2 Landing gear. The trailer shall be equipped with an adjustable, fully retractable landing gear with dual wheels and rubber tires. The landing gear shall support raise and lower the trailer at both curb and gross weights when disconnected from the prime mover and shall be fully retractable when the trailer is coupled to the prime mover. The landing gear shall remain positively locked when placed in either the raised or lowered position. When retracted and locked, adequate ground clearance shall be provided for cross country operations. The landing gear shall withstand the shock, vibrations and other treatments it may experience while being towed over cross country terrains without becoming loose from the locked position.

3.3.3 Suspension system. The axle, springs, accessories and components of the suspension system shall withstand the imposed loads without evidence of overload permanent set. Failure or malfunctioning under the operating conditions specified. The suspension shall be of the multileaf spring type suspension with shock absorbers. Clearance between the springs and spring stops shall be sufficient to prevent frequent bottoming. The trailer shall be equipped with a single axle and dual commercial type tires and wheels.

3.3.4 Wheels and tires. Wheels and tires shall comply with all requirement of the Tire and Rim Association for this type vehicle. Tires shall be of the steel belted radial type construction and shall have all terrain type tread. all wheels and tires on the trailer shall be identical.

3.3.5 Wheel housing. The trailer shall be equipped with wheel housings to protect the trailer frame, cargo, tank, pumping system fuel servicing hose personnel and towing tractor tires from road splash and stone throw. Wheel housings shall be of rugged construction,, adequately reinforced and provided with rounded corners and turned under lip to eliminate sharp edges. Sufficient clearance shall be provided for easy removal and replacement of the tire chains and tire wheel assemblies: tire chain clearance shall be in accordance with SAE J683a. Each wheel housing shall be capable of withstanding a static load of 250 pounds. Minimum fender construction shall be 10 gauge steel.

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3.3.5.1 Mud flaps. Wheel housings specified in 3.3.5 shall be provided with rubber mud flaps. These flaps shall provide protection to the rear against rear wheel splash and stone throw and shall comply with SAE J682.

3.3.6 Rear bumper. A heavy duty bumper in accordance with 49CFR 178.340-8(b) and 49CFR 393.86. shall be provided at the rear of the trailer. The outer edges of the bumper shall be rounded and shall incorporate other protection as required to minimize the hazard to operating personnel walking around the trailer. Lights and reflectors shall be recessed into the bumper in a protective manner.

3.3.7 Brakes

3.3.7.1 Service brakes. The service brakes of the trailer in conjunction with the towing vehicle under all conditions of loading shall control decelerate and stop the vehicle combination in accordance with 49CFR 393.52. The braking system shall comply with 49CFR 393.40 40CFR 393.42 49CFR 393.43 and 49CFR 393.45 through 49CFR 393.47. Service brakes shall hold the fully loaded trailer motionless and control it on a minimum 30 percent incline. Operation of the service brakes shall not result in evidence of excessive fading nor shall the vibrations and shocks encountered during cross country operation cause the brakes to lock. Application of the brakes on all wheels of the trailer and the towing vehicle shall be concurrent. The system shall be complete with a breakaway (spring applied) and emergency braking in accordance with 49CFR 393.43. The braking system shall be installed in a manner to provide adequate protection against debris encountered during off road use. The system shall be complete with air filters, air reservoir, slack adjusters, relay emergency valves and hose connectors fitted with dummy couplings.

3.3.7.2 Parking brakes. The trailer shall be equipped with a parking brake system in accordance with 49CFR 393.41. Parking brakes shall hold the trailer at rated capacity on a dry hard surface grade of not less than 30 percent when uncoupled from the towing vehicle resting on the running gear. The parking brakes shall locke the wheel sin order that they shall not roll but skid on a dry level paved surface when the fully loaded trailer is subjected to a towing force sufficient to move the vehicle.

3.3.7.3 parking brake override switch. A locking on/off parking brake override switch shall be provided. The switch shall be clearly marked. "CAUTION - PARKING BRAKE OVERRIDE - DO NOT USE EXCEPT TO POSITION TRAILER". The trailer air reservoir shall be have sufficient capacity to provide a minimum of five parking brake override cycles to allow for moving the vehicle when it is not connected to the tow vehicle.

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3.3.7.4 Gladhands, brake lines and fittings. Gladhands, brake lines and fittings shall be in accordance with SAEJ318.49CFR 393.45.49CFR 393.46. All brake lines and fittings shall be internally clean prior to the after making connections. The brake lines shall be securely anchored to the chassis in a manner to prevent chafing, kinking or other damage.

3.3.8 Frame. Frame rails shall have a minimum section modulus of 6.0 cubic inches each. Frame rails and crossmembers shall have a minimum yield strength of 36,000 psi. The contractor shall not weld to the frame assembly or add any holes to the frame rail flanges.

3.4 Lighting system. The trailer shall be equipped with a 12 volt lighting system compatible with commercial truck-tractors and a 24 volt lighting system compatible with military vehicles. Wiring, lamps, reflectors and associated signaling equipment shall conform to SAEJ1292 and MIL-STD-1179. Wiring shall be color coded or numbered for easy identification and shall be vaporproof enclosed in rigid metal conduit firmly attached to the trailer. All lights, reflectors and signaling equipment shall be protected from operational hazards by mounting in recessed or otherwise guarded locations.

3.5 Tank assembly

3.5.1 Tank assembly. The tank shall be of all welded construction, single compartment type with a 600 gallon minimum liquid capacity plus a minimum three percent expansion space. The tank shall have an elliptical or modified rectangular cross section. Tank design shall meet all applicable requirements of NFPA 385. NFPA 407 and MC-306. There shall be adequate vents, controls and piping for the trailer's intended purpose. The tank shall be designed to withstand loads imposed by operation on unimproved roads and cross country terrain. All components of the tank and piping which may come into contact with the liquid cargo shall be fabricated of aluminum or 300 series stainless steel. The minimum sheet thickness shall be 0.250 inches for at least the bottom third of the tank with the remaining minimum thickness of 0.187 inches. The interior of the tank shall have a bright clean finish water stained aluminum shall not be permitted. No welds shall be made outside the tank parallel and directly opposite to welds inside the tank. The tank shall be designed so that drainage from the vents and overflow shall drain towards the rear of the vehicle.

3.5.1.1 Filling rates. The tank shall be capable of being filled through the manhole filler cover at a rate of 200 GPM. Utilizing the trailers pumping system the trailer shall have the capability of drawing fuels from a storage tank four feet below the pump mount through a suction hose at a minimum rate of 20 GPM. During loading operations there shall be no excessive splashing due to interfering structure or other characteristics of the tank design.

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3.5.1.2 Emergency shutoff valve. Emergency shutoff valves and shear sections shall be provided for tank openings as required by NFPA 385.

3.5.2 Baffles. Baffles shall be designed, constructed and installed in accordance with NFPA 407 and 49CFR 340-7. The baffles shall be dished and flanged to absorb dynamic loading under all product load configurations. Ring stiffeners or other reinforcement designs will not be acceptable in lieu of baffles. The baffles shall be designed and constructed to permit inspection and cleaning of the entire tank. A minimum of two baffles nor stand in pockets.

3.5.3 Sump. A sump having a minimum capacity of 12 gallons shall be provided at the bottom of the tank. The tank shall be designed to ensure that all water will drain into the sump and that water will neither be trapped by baffles nor stand in pockets.

3.5.3.1 Sump drain. A one inch drain valve shall be installed in the sump to permit complete drainage of the tank and sump. The valve shall be protected against foreign objects and shall be easily accessible. The outlet of the drain shall be directed to the ground.

3.5.4 Manhole. The tank shall have a commercial circular manhole with a minimum diameter of 20 inches. The manhole shall have a watertight 10 inch filler cover with necessary safety features to eliminate the possibility of a spark occurring during opening or closing of the cover. The manhole shall be equipped with a Buna N gasket or equivalent to insure a fuel tight seal between the collar and the cover and shall maintain an internal pressure of 3 psig without leaking. The entire top of the manhole shall be removable by removing a bolt on clamping ring. Access to the manhole shall be provided in accordance with MIL-STD-1472.

3.5.5 Vents. The tank shall be equipped with two identical removable vents. The vents shall not open until the pressure or vacuum in the tank reaches 0.2 psig and shall not permit the tank pressure to exceed 2 psig nor the tank vacuum to exceed 1 psig. The vents shall remain closed at all other times. The vents shall be designed to prevent loss of fuel in the event the trailer overturns and shall be located and protected in a manner to prevent water, snow, wind, blown sand, etc. from entering the tank when the vent is open.

3.5.6 Tank mounting. The tank mounting brackets shall be attached plates extending at least the width of the frame rails. The plates shall be a minimum 0.220 inch thick aluminum at least 12 inches wide, and all edges of the plates shall be completely welded to the tank. The tank shall be mounted on rubber pads; there shall be no metal to metal contact between the from rails and tank mounting hardware.

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3.6 Pumping system. The pumping system shall be capable of performing the pumping requirements specified herein when delivering fuels conforming to VV-F-800, MIL-G-3056, MIL-G-5572, MIL-T-5624, MIL-T-38219 and MIL-T-83133. The trailer shall be equipped with an engine driven pump capable of delivering fuel flow adjustable from 0 GPM to 20 GPM through the servicing hose and nozzle to a height of 20 feet above the base of the pump and shall have the capability of drawing fuel from a level four feet below the base of the pump. The pumping system shall also be capable of pumping fuel from a separate container through two 10 foot suction hoses. The pump shall be in accordance with 49 CFR 178.340-9.

3.6.1 Pumping compartment. The pumping system and all its components unless otherwise specified herein shall be housed in a completely enclosed compartment located in the front of the cargo tank. The compartment shall be completely removable and shall be capable of being removed and reinstalled within five minutes using common hand tools. The compartment shall have a full width vertical hinged compartment door located on each side of the trailer. The hinged door shall have a full length piano type hinge with a stainless steel pin and shall be attached with threaded fasteners. Fasteners shall be recessed or flush with compartment interior panels. The doors shall have provisions for adjustment to ensure smooth easy operations. When in the open position, convenient access shall be provided to all components of the pumping system. Each door shall be equipped with a handle latch and automatic hold open device. The constructed of stainless steel or aluminum. The assembly shall incorporate necessary structural integrity to prevent bending, warpage or other type of damage due to vibration and other forces encountered during normal operation. A one inch plugged drain hole with a removable drain plug shall be provided at the lowest point of the compartment to permit complete drainage of excess fuel accumulated on the compartment floor. The drain plug shall be securely attached to the compartment. The compartment shall allow the fuel servicing hose specified in 3.7.4.1 to be attached to the discharge stub without the opening of a compartment door.

3.6.2 Engine. The pump shall be driven by a commercial air cooled, hand cranked, diesel engine. Engine torque and horsepower shall be produced at governed speed which will assure performance with an adequate margin of safety. The total power output required during any pumping operation specified herein shall not exceed its net power output rating at 2000 feet elevation, determined in accordance with SAE J1349. The engine shall be capable of providing the performance required herein when operating on diesel fuel conforming to VV-F-800, Grades JP-4 and JP-5 turbine fuel conforming to MIL-T-5624 and Grade JP-8 turbine fuel conforming to MIL-T-83133. The engine exhaust shall be vented downward outside the pumping compartment and shall be equipped with a spark arrestor. An emergency cut off

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device shall be provided and located in an easily visible and accessible location for quick response. The fuel tank shall have sufficient capacity to allow at least five hours of pumping at maximum flow rate. All hoses associated with the engine and fuel tank shall be fitted with hose clamps in accordance with SAE J536b, Types B, D or F (Style 4).

3.6.3 Hand pump assembly. In addition to the engine drive pump assembly, a hand pump shall be provided.

3.7 Gauges

3.7.1 Pressure gauge. A pump discharge pressure gauge shall be provided to continuously monitor pressure at the outlet of the engine driven pump. The gauge shall be at least three inches in diameter with at least a 0 psi to 150 psi scale in 1 psi graduations. The gauge shall have a plus or minus 1.5 psi accuracy over the full scale range and shall incorporated a restriction plug to dampen oscillations and to provide stability of the operating needle. The gauge shall be marked with a red line as to the maximum safe operating pressure. The gauge shall have a minimum face diameter of three inches.

3.7.2 Flowmeter. A flowmeter of at least 25 GPM capacity shall be furnished. The meter shall withstand the pumping system hydrostatic pressures without leakage or damage and shall be installed so that all fuel dispensed through the fuel servicing hose or drawn through the suction hose is metered. The meter shall be clearly visible and within easy reach of an operator standing at the engine control panel. The meter shall have a shatterproof lens.

3.7.3 Recirculation system. The trailer's pumping system shall incorporate a recirculation circuit which is capable of returning the maximum flow rate of fuel from the pump output to the cargo tank when the pump is operated with the hose nozzle closed.

3.7.4 Hose and nozzle

3.7.4.1 Hose. A 35 foot long 1 1/2 inch inside diameter fuel servicing hose conforming to MIL-H-6615 shall be attached to the vehicle pumping system. Two 10 foot lengths of two inch inside diameter hose conforming to MIL-H-6615 shall be provided in a hose receptacle assembly. The two hoses shall be available for connecting to the pumping suction system for dedrumming operations.

3.7.4.1.1 Hose couplings. Each of the suction hoses shall be equipped with male fittings and female fittings conforming to MS 27020-11 and MS 27026-11 on opposite ends. Each fitting shall be complete with a dust cap conforming to MS 27028-11 or MS 27029-11. The servicing hose shall be equipped with a male adapter, MS 27022-9 and dust cover, MS 27028-9.

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3.7.4.2 Hose storage. A rack for securely storing the fuel servicing hose shall be furnished and attached to the cargo tank or frame assembly. The fuel servicing hose storage shall be on the same side as the discharge outlet and shall allow the hose to remain attached to the discharge stub. The storage rack shall securely stow the hose under all towing operations. Two aluminum cylinders with covers shall be attached to the trailer frame for storing the two 10 foot suction hoses. The covers shall be securely attached to each cylinder.

3.7.4.3 Nozzle. A one inch automatic shutoff nozzle. OPW Part Number OPW-7H or equal shall be furnished and attached to the fuel servicing hose with a reducer. The nozzle shall be equipped with a removable dust cap securely attached so that it cannot become separated from the nozzle. The nozzle shall have a quick release clamp or other hardware to securely stow the nozzle when the fuel servicing hose is in its stowed position.

3.8 Piping

3.8.1 Piping design. The piping sections shall be designed to ensure electrical continuity across each connection. Flexible couplings shall be provided between the tank and pumping sections. All piping shall be interchangeable without special tools. Failure in piping shall not occur as a result of road operation or torsional forces.

3.8.2 Line strainer. A 100 mesh strainer shall be incorporated into the trailer pumping system. The strainer shall be readily accessible for ease of cleaning.

3.8.3 Water segregator. The suction line shall be equipped with a water segregator which will shut off the flow automatically when the water level in the tank reaches the capacity of the sump. The suction line shall be designed so water cannot be drawn from the tank sump through the suction line.

3.8.4 Suction and discharge stubs. A female e adapter MS 27024-9 shall be provided in the pump discharge piping for attaching the servicing hose. The adapter shall be equipped with a female adapter (MS 27024-11) and dust cover (MS 27029-11).

3.8.5 Shutoff valves. Pumping system shutoff valves of the ball type shall be provided as necessary to isolate components for maintenance or different

3.8.6 Relief valves. Pressure relief valves shall be installed in the piping system to relieve excessive pressure in any line due to system malfunction or thermal expansion of fuel trapped in the line. The relief pressure shall return to the cargo tank.

3.9 Accessories

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3.9.1 Fire extinguisher. One Halon 1211 fire extinguisher. Amerex Part Number 361 .or equivalent. shall be mounted in a location readily accessible to persons standing on the ground. The mounting location shall be such that the extinguisher will be protected from water, mud, slush, etc.. thrown up by the wheels.

3.9.2 Static cable reels. Two static cable reels conforming to MIL-R-83232/2 shall be installed in the pumping compartment. The location shall permit easy withdrawal and recovery of the grounding wires. The reels shall be so installed that the resistance between the reel and the frame does not exceed 0.5 OHM.

3.10 Finish and protective coatings.

3.10.1 Cleaning, painting, plating, anodic films and chemical treatments. Cleaning, painting, plating, anodic films and chemical treatments shall be in accordance with MIL-STD-808. Unless otherwise approved by the procuring activity, primer shall be in accordance with MIL-P-23377.

3.10.2 Finish

3.10.2.1 Forest green. Unless otherwise specified (see 6.2) the outer surface of the trailer and the interior surfaces of the compartment shall be painted with polyurethane paint in accordance with MIL-C-83286. Forest Green Color Number 24052 of FED-STD-595. All exterior trim items (such as wheels. wheel covers, and door handles) shall be the same color as the exterior finish of the trailer.

3.10.2.2 Desert sand. When specified (see 6.2), the outer surface of the trailer and the interior surfaces of the compartment shall be painted with polyurethane paint in accordance with MIL-C-46168 or MIL-C-53039, Desert Sand, Color Number 30313 of FED-STD-595. All exterior trim items (such as wheels, wheel covers, bumpers and door handles) shall match the exterior color of the trailer.

3.10.3 Rustproofing. Unless otherwise specified (see 6.2), the vehicle shall be rustproofed in accordance with the contractor's standard commercial practice. The first article units shall not be rustproofed until after approval of the first article test report.

3.11 Markings. Unless otherwise specified markings for vehicles painted Forest Green shall be applied with black paint, Color Number 37038 of FED-STD-595. and markings for vehicles painted Desert Sand shall be applied with white paint. Color Number 37835 of FED-STD-595.

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3.11.1 Identification markings. Unless otherwise specified (see 6.2), the vehicle shall be marked for the appropriate service in accordance with MIL-STD-1223.

3.12 Identification plates and operating instructions. All identification plates, instruction plates, data plates, etc shall conform to MIL-P-514 and be of a size and shape consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12.

3.12.1 Identification plate. An identification plate, permanently marked in accordance with MIL-STD-130, shall be securely attached to the vehicle in a readily accessible location. The identification plate shall contain the following information.

NOMENCLATURE

MANUFACTURER'S MODEL NUMBER

MANUFACTURER'S SERIAL NUMBER

MANUFACTURER'S NAME

MANUFACTURER'S ADDRESS

MANUFACTURER'S SERVICE PHONE NUMBER

DATE OF MANUFACTURE

CONTRACT NUMBER

NATIONAL STOCK NUMBER

3.12.2 Control markings. All controls, valves, gauges and indicators used in the operation of the vehicle shall be identified by securely attached nameplates.

3.12.3 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed near the appropriate controls. The instructions shall be clear, concise and adequate to enable operation of the vehicle without damage to the equipment or injury to personnel and shall refer to the components as identified by nameplates.

3.12.4 Lubrication plate. A lubrication plate shall be provided directing attention to all lubrication fittings and components which require lubrication. The plate shall identify the type and grade of lubricant required for all operational temperatures.

3.12.5 Transportation data plate. A transportation data plate shall be provided. The plate shall contain at least the following information:

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a. Side and rear silhouette views of the vehicle.

b. Horizontal and vertical location of the center-of-gravity of the vehicle in transport condition.

c. Shipping weight.

d. Loading cubage.

e. Overall length, width and height.

f. Tiedown instructions for air transport in Types C-130, C-141 and C-5 Aircraft.

3.13 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.14 Dimensions. The vehicle in fully operational condition including all equipment and accessories shall not exceed the following dimensions:

a. Length, including towbar	175 inches.
b. Width	96 inches.
c. Height	85 inches.

3.15 Welders and welding

3.15.1 Welders. All welders employed in the fabrication of the vehicle except for the commercial chassis shall be certified in accordance with the applicable requirements of MIL-STD-1595 the American Welding Society or the American Society of Mechanical Engineers.

3.15.2 Welding. The surface parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test proof and service loadings.

3.15.3 Screw assemblies. Unless torque is specified threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.15.4 Riveting. Riveting operations shall be performed to ensure tight fit and complete heading of the rivets.

3.15.5 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots or other irregularities. Where required for accurate adjustment gear assemblies shall be free of backlash.

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3.15.6 Cleaning. The vehicle shall be thoroughly cleaned, Loose, spattered or excess solder. Welding slag, stray bolts, nuts and washers, rust metal particles, pipe compound and other foreign matter shall be removed during and after final assembly.

3.16 Workmanship. The vehicle including all parts and accessories shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp corners and edges: accuracy of dimensions, surface finish and radii of fillets; thoroughness of soldering, welding, brazing, painting, wiring and riveting marking of parts and assemblies alignment of parts and tightness of assembly fasteners; etc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use any inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that the vehicle conforms to the prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the vehicle shall be classified as follows:

a. First article testing (see 4.4).

b. Acceptance testing (see 4.5).

4.3 Test conditions

4.3.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at proper intervals to ensure laboratory accuracy.

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4.3.2 Data. During all testing specified herein at least the following data unless not applicable shall be recorded at intervals not to exceed 15 minutes. Additional data and/or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time Finished.

d. Ambient temperature.

e. Engine speed.

f. Amount of fuel in GPM pumped.

g. Gauge readings.

4.3.3 Preparation for tests. The vehicle equipped with all specified auxiliary equipment shall be fully serviced and all equipment shall be prepared for immediate operation.

4.3.4 Test rejection criteria. Throughout all tests specified herein the vehicle shall be closely observed for the following conditions which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein.

b. Spillage or leakage of any liquid including fuel, coolant or lubricants under any operating condition except as allowed herein.

c. Structural failure of any component including permanent deformation or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between the vehicle components or between the vehicle, the ground and all required obstacles with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering, braking and while traversing rough terrain.

h. Conditions which present a safety hazard to personnel during operation, service or maintenance.

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i. Overheating of the engine or any other vehicle component.

4.4 First article testing

4.4.1 First article test sample. The first article test sample shall be a preproduction item which meets the requirements of this specification. The sample shall be identical to the items procured under the terms of the contract. Approval of the first article samples shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.4.2 First article tests. The first article tests shall consist of all test listed under 4.6.

4.4.3 Test report. Upon completion of the first article tests a first article test report shall be prepared in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets and the report. The following shall be incorporated as a part of the test report:

a. All failures maintenance performed and other events recorded shall be identified by accumulated operation time miles, cycles or position of the test procedure as appropriate. Test conditions at the time of failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made or proposed to be made and the predicted effectiveness of those corrections.

d. Test activity and/or contractor comments on item features or requirements that if modified should improve the item.

e. Test activity and/or contractor comments on item features or requirements that if modified should improve the item.

f. List of all cleaning and finishing procedures to be used in accordance with MIL-STD-808.

g. List of methods to be used to protect against electrolytic corrosion of dissimilar metals in intimate contact in accordance with MIL-STD-889.

h. Test report for each component covered by a document listed in the Department of Defense Index of Specifications and Standards (DoDISS).

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i. Test report for each component or system covered by a publication listed in this purchase description (i.e., NFPA. CFR and SAE publications).

j. Plan for testing production units.

4.5 Acceptance testing. Acceptance testing shall be classified as follows:

a. Individual testing (see 4.5.1).

b. Random sample testing (see 4.5.2).

4.5.1 Individual testing. Each vehicle may be subjected to the tests specified in 4.6.1.4.6.2.2.4.6.2.3.4.6.2.4.4.6.2.5.4.6.2.7.1.4.6.2.7.2 and 4.6.2.8.

4.5.2 Random sample testing. Any production vehicle may be selected by the procuring activity at any time to be subjected to the test described in 4.6. Upon completion of the random sample testing the contractor shall prepare a test report in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets of the report.

4.5.2.1 Rejection and retest. When a vehicle fails to meet the requirements of this specification vehicles to be delivered pursuant to the contract may not be accepted until the cause of the failure is determined corrective action is taken by the contractor and the component that failed is successfully retested to assure compliance with this specification. Tests not involving the failed component may be continued pending the investigation of a test failure.

4.6 Test methods

4.6.1 Examination of product. The vehicle shall be inspected to determine compliance with the requirements specified herein. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and their application, welding and riveting and markings. For the first article and random sample vehicles. This examination shall be accomplished by use of a checklist which list each requirement not validated by a test and the results of the examination.

4.6.2 Preliminary operation

4.6.2.1 Test fluids. Unless otherwise specified herein the test fluids for first article testing shall be JP-4 in accordance with MIL-T-5624 for the pump engine and DF-2 diesel fuel oil in accordance with VV-F-800 for the cargo tank.

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4.6.2.2 Static pressure test. The cargo tank shall be subjected to an internal static air pressure of 3 psig for not less than 30 minutes. A soap solution shall be sprayed on the exterior of the tank to facilitate detection of leaks which may exist. Evidence of leakage shall be cause for rejection.

4.6.2.3 Hydrostatic pressure test. The flow system and components of the pump assembly on the pressure side of the pump mounted on the trailer shall be subjected to 125 psig for not less than 30 minutes. There shall be no leaks in the system.

4.6.2.4 Vent operation test. The venting equipment shall be removed from the tank and installed in a test fixture to verify conformance with 3.5.5. The vents shall relieve positive and negative pressures and prevent spillage or leakage of fuel in any direction.

4.6.2.5 Pumping operations.

4.6.2.5.1 Hydrodynamic loading. The tank shall be filled through the filler opening at a minimum rate of 200 gpm through a two inch diameter pipe. There shall be no excessive splashing due to interfering structure or other characteristics of the tank design. There shall be no evidence of damage or permanent distortion due to high flow rates.

4.6.2.5.2 Drawing test. The trailer shall pump fuel from a storage tank level four feet below the pump mount through one section of suction hose at a minimum rate of 20 gpm. The test shall be conducted with the nozzle at the level of the pump outlet. A minimum of 600 gallons shall be pumped.

4.6.2.5.3 Elevation pump test. The cargo tank shall be filled to capacity and the fuel servicing icing hose shall be elevated a minimum 20 feet above the base of the pump. A flow rate of 20 gpm shall be established and a minimum of 500 gallons shall be pumped. There shall be no leaks in the pumping system.

4.6.2.5.4 Pumping system test. The cargo tank shall be filled to capacity. For first article testing the hose nozzle shall be placed in the tank fill opening and a minimum of 5,000 gallons of fuel shall be pumped at maximum engine speed. During the test the nozzle shall be closed not less than tree times for not less than 30 seconds each time. The engine shall be run at least five hours before refueling is allowed. Average fuel consumption shall be calculated and recorded in the test report. During testing the automatic nozzle shutoff feature shall be demonstrated 50 times. For acceptance testing a minimum of 500 gallons of fuel shall be pumped at a maximum engine speed. There shall be no leaks in the pumping system.

4.6.2.5.5 Hand pump test. With the tank filled to capacity the hose nozzle shall be placed in the tank fill opening and a

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minimum of 50 gallons of fuel shall be pumped with the hand pump.

4.6.2.6 Pumping system enclosure. The pumping system enclosure shall be removed and replaced to demonstrate compliance with 3.6.1.

4.6.2.7 Roadability tests. Unless otherwise specified the following tests shall be conducted using a military design 2 1/2 ton cargo M35 vehicle.

4.6.2.7.1 Road handling. The trailer shall be filled to design capacity and towed not less than five miles over paved highways or similar improved roads and not less than two miles over graded gravel or similar unimproved roads. Throughout the road handling test the trailer shall be observed as to its trackability, ease of handling, tendency to yaw, tendency to sway, etc. The path of the trailer shall not deviate more than three inches to either side of the path of the prime mover. Upon completion of the road handling test the trailer shall be inspected for misalignment, distortion or other damage sustained as a result of the testing. Undesirable roadability features, interference or any damage resulting from the test th at would affect its operational use shall be cause for rejection. A complete description of any adverse handling characteristics or damage sustained shall be recorded.

4.6.2.7.2 Turning ability. The trailer shall be filled to design capacity and coupled to the truck tractor. The vehicle combination shall be operated through right and left turns, both towing and backing in which the towing vehicle assumes angles from 0° to 70° with the trailer. A minimum of ten right and ten left turns shall be conducted each at different turn configurations. In addition the applicable turning abilities in accordance with SAE J695 shall be measured and recorded. Throughout the turn ability testing the trailer shall be checked for sufficient clearance with the prime mover. Upon completion, the trailer shall be inspected for misalignment, distortion or other damage sustained as a result of the testing. Undesirable turnability features and any damage resulting from the test that would affect its use and operation shall be cause for rejection.

4.6.2.7.3 Slope test. The trailer shall be operated on specified slopes and observed for compliance to 3.2.1a. With a 600 gallon load, the trailer shall be capable of operating satisfactorily on a 30 percent maximum side slope without leakage of fuel through the vents. Under the same load condition the trailer shall be capable of operating satisfactorily on a 30 percent maximum longitudinal slope without leakage of fuel through the vents.

4.6.2.8 Braking tests. The service brakes shall be tested to demonstrate compliance to 49 cfr 393.52.

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4.6.2.1 Parking brake override test. The trailer shall be parked on a hard level surface the landing gear lowered and the towing vehicle disconnected. With the parking brake override switch in the OVERRIDE position it shall be demonstrated that the trailer rolls freely. The parking brake is so to be reactivated and four additional override cycles performed.

4.6.2.9 Mobility tests. The trailer with JP-5 fuel as the test fluid and when coupled to the 2 1/2 ton M35 vehicle shall be subjected to the following test plan:

a. At rated capacity -one cycle.

b. 40 to 60 percent rated capacity - one cycle.

c. Empty - one cycle.

Each cycle shall consist of the following phases:

a. Highway - The trailer shall be towed over hard surface roads at speeds ranging from 20 to 50 MPH with an average minimum speed of 40 MPH for a minimum of 200 miles. Not less than five maximum stops, ten right hand and ten left hand (70°) turns shall be performed.

b. Off Highway (Cross Country) - The trailer shall be towed for a minimum of 250 miles over cross country terrain consisting of unimproved roads, open fields, broken ground, loose sand and gravel roads such as encountered by tactical vehicles. The distance towed over each type of terrain shall be approximately equal. Vehicle speed shall be at least 10 MPH.

c. Belgian Block - The trailer shall be towed over Belgian Block for 250 miles at an average minimum speed of 10 MPH. It shall be demonstrated that the vehicle can be towed to demonstrate compliance with 3.2.1f. An approved cobblestone course may be used as an alternate to Belgian Block.

d. Pump operation - If empty, the tank shall be filled to 50 percent rated capacity. The landing gear shall be lowered and the trailer disconnected from the towing vehicle. The hose nozzle shall be placed in the tank fill opening and the pump engine stared. A minimum of 500 gallons of fuel shall be pumped at minimum flow rate of 20 GPM. The trailer shall be reconnected to the towing vehicle and the landing gear raised. The ability of the landing gear to raise and lower the vehicle shall be demonstrated. Throughout the mobility tests, the trailer shall be observed as to its trackability, ease of handling, tendency to yaw, tendency to sway, etc. Undesirable roadability characteristics difficulty of operation, binding of any component or failure of any component to perform as required by Section 3 of this purchase description shall be cause for rejection.

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4.6.2.10 Strain gauge testing. The following strain gauge tests shall be performed on the trailer.

4.6.2.10.1 Test course. For each test cycle the trailer shall be towed for a minimum of 10 miles over cross country terrain consisting of unimproved roads, open fields, broken ground, loose sand and gravel roads such as encountered by tactical vehicles. The distance towed over each type of terrain shall be approximately equal.

4.6.2.10.2 Patterning. High stress areas of the vehicle frame assembly, tank and tank mounting system shall be coated with brittle lacquer. The cargo tank shall be filled to approximately 50 percent rated capacity with JP-5 fuel and the trailer towed over the test course at a minimum speed of 5 MPH.

4.6.2.10.3 Strain recording. Strain gauges shall be positioned as indicated by the brittle lacquer patterning. A minimum of 10 strain gauges shall be used. Strains shall be recorded for each cycle of the test course:

a. At rated capacity - one cycle.

b. 40 to 60 percent rated capacity - one cycle.

c. Empty - one cycle.

Strain/occurrence histograms shall be supplied for each gauge for each test cycle. The actual mileage for the cycle shall be recorded on the histogram. A picture of each gauge indicating its position shall be provided.

4.6.2.10.4 Evaluation. The minimum ultimate tensile and yield strength of the gauged components shall be documented. Any strain recorded indicating stress greater than the material minimum yield strength shall be cause for rejection.

Fatigue life calculations shall project a vehicle life in excess of 10,000 miles under the following load conditions:

a. At rated capacity - 2,500 miles.

b. 40 to 60 percent rated capacity - 5,000 miles.

c. Empty - 2,500 miles.

4.6.2.11 Design verification

4.6.2.11.1 Mechanical operation. A thorough check shall be made of the trailer to ensure proper functioning of all mechanical parts, accessibility to all controls and levers, interchangeability of removable parts installation of equipment without fitting and to effect every degree of safety and performance.

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4.6.2.11.2 Wheel housing loading. It shall be demonstrated that each wheel housing can withstand a static load of 250 pounds. The load shall be placed on the outermost edge at three different locations each location shall support 250 pounds.

4.6.2.11.3 Weight and weight distribution. The weight on each corner of the trailer shall be measured with the unit on a flat level surface, both in the empty and the fully loaded conditions.

4.6.2.11.4 Dimensional verification. The dimensions specified herein shall be measured with the trailer standing on a flat level surface. Overall height and body lean shall be measured in both the empty and the fully loaded conditions.

4.6.2.11.5 Sound level testing. The trailer shall be tested to demonstrate compliance with 3.1.8.

4.6.2.11.6 Electromagnetic interference measurement. The vehicle and all electrical components shall be tested in accordance with MIL-STD-462 to demonstrate compliance with 3.1.7.

4.6.2.12 Air transportability analysis. A detailed engineering analysis shall be conducted to demonstrate compliance with 3.1.9. The analysis shall include the tiedowns and all major components and their ability to withstand the accelerations specified in MIL-STD-1791. The evaluation shall also include a dimensional analysis for the vehicle while loaded aboard a Type C-130 Aircraft.

4.6.2.13 Ramp test. The fully loaded vehicle shall be driven forward from a hard, flat, horizontal surface and ascend a ramp with a minimum length of 1.2 times the vehicle and a slope of 30 percent. It shall then proceed across the ramp crest onto an upper connecting hard horizontal surface until the tires are resting on the upper surface down the ramp and onto the lower horizontal surface. The test shall then be repeated in the opposite direction. During the test there shall be no evidence of interference or cramping between the prime mover and the trailer.

4.6.3 Environmental test

4.6.3.1 Low temperature storage and operation. The complete vehicle including all accessories shall be tested in accordance with Test Method 502.2. Procedures I and II, of MIL-STD-810 to demonstrate compliance with 3.1.4 (a). The test fuel for the pump engine and cargo tank shall be DF-2 diesel fuel in accordance with VV-F-800. Temperature sensor locations shall be approved by the procuring activity. "Standard ambient" and "controlled ambient" shall be defined as the ambient condition of the test site.

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For Procedure I, the vehicle shall be exposed to the minimum required cold storage temperature of 3.1.4(a) in a steady-state condition for 24 hours. For Procedure II, the vehicle shall be exposed to the minimum required cold operational temperature of 3.1.4(a) in a steady-state condition.

4.6.3.2 High temperature storage and operation. The complete vehicle including all accessories shall be tested in accordance with Test Method 501.2. Procedures I and II of MIL-STD-810 to demonstrate compliance with 3.1.4(a). The test fuel for the pump engine shall be JP-4 fuel and the test fuel for the cargo tank shall be JP-5 fuel in accordance with MIL-T-5624. Temperature sensor locations shall be approved by the procuring activity. "Standard ambient" and "controlled ambient" shall be defined as the ambient condition of the test site. For Procedure I the vehicle shall be exposed to the maximum required hot storage temperature of 3.1.4(a) in a steady-state condition for 24 hours. For Procedure II the vehicle shall be exposed to the maximum required hot operational temperature of 3.1.4(a) in a steady-state condition.

4.6.3.3 Rain. The complete vehicle including all accessories shall be tested in accordance with Test Method 506.2 Procedure III, of MIL-STD-810 to demonstrate compliance with 3.1.4(c). The interior of the compartment shall be examined for water any evidence of leakage shall be cause for rejection.

4.6.3.4 Other environmental tests. The components and materials used in the construction of the vehicle shall be subjected to or certified as having passed or being capable of passing the following test methods of MIL-STD-810 to demonstrate compliance with 3.1.4.

a. Humidity - Method 507.2 Procedure I.

b. Salt Fog - Method 509.2 Procedure I.

c. Sand and Dust - Method 510.2 Procedures I and II. For Procedure I. air velocity shall be 1-50 ft/min and high temperature adjustment shall be 125°. For Procedure II. air velocity shall be 5700 ft min: sand concentration of 0.0623 = 0.015 g/ft^3; 90 minutes per face duration.

4.6.4 Maintainability demonstration. Compliance with the maintenance downtime requirements shall be demonstrated by task performance. All scheduled maintenance tasks and all tasks listed in 3.1.3.2 shall be performed. In addition corrective maintenance tasks shall be selected by the procuring activity from an approved list of tasks provided by the contractor in the first article test procedure. Not less than 25 percent or more than 50 percent of the total number of tasks shall be demonstrated to verify an average maintenance downtime within the required limit. As part of this demonstration the

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recommended frequencies of the scheduled maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of scheduled maintenance downtime per measure of use such as calendar time hours of operation or miles traveled. The capability of performing maintenance tasks by personnel wearing arctic mittens shall be demonstrated.

4.6.5 Structural analysis. Stresses shall be calculated for worst cause conditions of forces combined in phase. Maximum stress shall not exceed the component minimum yield strength: a Goodman analysis of all fasteners with maximum installation torque shall indicated infinite life (500,000,000 cycles for nonferrous fasteners).

4.7 Refurbishment. After successful completion of all required testing each vehicle shall be restored to like new condition.

5. PACKAGING

5.1 Packaging. Shall be in accordance with the terms of the contract.

6 NOTES

6.1 Intended use. The Type A1-B trailer covered by this purchase description is intended for use in fuel servicing all types of aircraft, ground and ground support equipment.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Finish color required if not Forest Green. Color Number 24052 of FED-STD-595 (see 3.10.2.2).

c. Identification markings required (see 3.11.1).

d. Rustproofing, if other than the manufacturer's standard commercial practice is required (see 3.10.3).

6.3 Definitions

6.3.1 Belgian Block. A belgian Block course is paved with unevenly laid granite block forming an undulating surface. It duplicates a rough cobblestone road such as is found in many parts of the world.

6.3.2 Cross country terrain. Cross country terrain consists of open fields, broken ground and uneven terrain in general.

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6.4 Subject term (Key Word) listing

Forced Discharge
Fuel Servicing
Rough Terrain
Trailer, Tank

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 84

Project Number:
2330-F988

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DEPARTMENT OF THE AIR FORCE

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
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POSTAGE WILL BE PAID BY
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-87984	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*,
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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